

Zen Nguyen

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Hiring Manager
BrickRed Systems

Re: Software Engineer Intern - (GCP / Firebase / AI Tools)

I am incredibly excited about the Software Engineer Intern role at BrickRed Systems, particularly the opportunity to contribute to building modern web applications and AI-enabled solutions using GCP and Firebase. My personal project, TitanSwarm, directly aligns with this focus, where I architected an autonomous AI Co-Pilot in Python utilizing LangChain, Gemini API, and a FAISS Vector Store to automate job application workflows. This project involved building a zero-hallucination RAG engine and a Streamlit UI, demonstrating my ability to work with AI tools and develop end-to-end solutions.

Furthermore, my experience with SFU Course Tracker showcases my proficiency in full-stack development with TypeScript and FastAPI, including migrating to AWS and implementing Docker container orchestration. This project also involved designing and implementing a custom parser for complex boolean strings and engineering a high-throughput scraping pipeline, skills that are directly applicable to building scalable web applications. My work on TitanStore, a distributed key-value database built in Go using the Raft consensus algorithm and TCP sockets, highlights my understanding of backend systems and fault-tolerant architecture, which are crucial for cloud-native development.

I am eager to apply my skills in TypeScript, AI tools, and cloud technologies to the challenges at BrickRed Systems and contribute to your innovative projects. Thank you for considering my application; I look forward to discussing how my background can benefit your team.

Sincerely,

Zen Nguyen